

Draw **the tangent** AD , the straight line that just grazes **the curve** at A , matching its direction there and extending both ways. Between **the tangent** and **the chord** opens **the angle** BAD . As B approaches A , **the chord** swivels toward **the tangent**, and **this angle** is diminished *in infinitum*. For if **the angle** did *not* vanish, **the arc** would meet **the tangent** at a finite rectilinear angle — a sharp elbow at A — and the curvature there would not be continuous, against the supposition. Hence ultimately $\angle BAD \rightarrow 0$. *Q.E.D.*